

IT Firm Assessment Project

Employee Management & Analytics System

Assessment Information

Project Name	Employee Management & Analytics System
Difficulty	Intermediate (IT Company Level)
Duration	8 Hours
Technologies	Python, NumPy, Pandas, CSV, JSON, Pickle
Database	Not Allowed

Business Scenario

ABC Technologies has approximately 5,000 employees.

Currently, HR manages employee records manually using Microsoft Excel. Your responsibility is to develop a Python application that automates:

- Employee Management
- Salary Analytics
- Performance Analysis
- Department Reports
- File Storage
- Error Handling

1 Module 1 : Python Foundations

Task 1 : Variables and Data Types

Create variables for:

- Company Name
- Employee ID
- Employee Name
- Department
- Salary
- Experience
- Performance Rating
- Joining Date

Task 2 : Operators

Use

- Arithmetic Operators
- Comparison Operators
- Logical Operators

Determine

- Promotion Eligibility
- Bonus Eligibility
- Salary Increment

Task 3 : Collections

Use

- List
- Tuple
- Set
- Dictionary

Store employee information.

Task 4 : Control Flow

Use

- if
- elif
- else

Classify employees as

- Excellent
- Good
- Average
- Poor

Task 5 : Loops

Generate

- Employee IDs
- Salary Report
- Attendance Summary

Task 6 : List Comprehension

Find employees satisfying

- Salary > 50000
- Experience > 5 Years

2 Module 2 : Functions and Modules

Task 7

Create functions

- add_employee()
- update_employee()
- delete_employee()
- search_employee()
- calculate_bonus()
- calculate_tax()
- promotion_check()

Task 8

Use

- *args
- **kwargs

for flexible employee creation.

Task 9

Use

- lambda
- map()
- filter()

for salary processing.

Task 10

Using datetime module,
Calculate employee experience.

Task 11

Using os module,
Automatically create

- Reports
- Backup
- Logs

folders.

Task 12

Using Regular Expressions,
Validate

- Email
- Phone Number
- Employee Code

3 Module 3 : Object Oriented Programming

Task 13

Create Employee class

Attributes

- ID
- Name
- Department
- Salary
- Experience
- Performance

Methods

- display()
- bonus()
- promotion()
- tax()

Task 14

Create subclasses

- Manager
- Developer
- Analyst

using Inheritance.

Task 15

Use Encapsulation with private variables.

Task 16

Use @property for salary validation.

Task 17

Use @classmethod to count employees.

Task 18

Use @staticmethod to calculate PF.

4 Module 4 : File Handling

Task 19

Store employee records in CSV.

Task 20

Store employee records in JSON.

Task 21

Store Employee objects using Pickle.

Task 22

Read all files.

Task 23

Update employee records.

Task 24

Delete employee records.

5 Module 5 : Exception Handling

Handle

- Invalid Salary
- Negative Experience
- Duplicate Employee ID
- File Not Found
- Division by Zero
- Invalid Menu Choice

Task 25

Create custom exception
EmployeeNotFound

Task 26

Use finally block for logging.

Task 27

Use Context Managers
(with statement)
for every file operation.

6 Module 6 : NumPy

Create

- Salary Array
- Experience Array
- Performance Array

Task 28

Perform

- Shape
- Dtype
- Reshape
- Flatten
- Broadcasting

Task 29

Use

- `sqrt()`
- `log()`
- `exp()`

Task 30

Use `np.where()`
Generate

- Bonus
- Promotion
- Salary Grade

Task 31

Find

- Mean
- Median
- Variance
- Standard Deviation
- Maximum
- Minimum

Task 32

Perform

- Dot Product
- Matrix Multiplication
- Transpose
- Determinant
- Inverse

7 Module 7 : Pandas

Task 33

Import data from

- CSV
- Excel
- JSON

Task 34

Inspect Dataset

- shape
- columns
- dtypes
- info()
- describe()

Task 35

Handle Missing Values

- `fillna()`
- `dropna()`

Task 36

Remove duplicate records.

Task 37

Convert data types using `astype()`.

8 Module 8 : Advanced Pandas

Task 38

Department-wise analysis using
`groupby()`

Task 39

Aggregate statistics using
`agg()`

Task 40

Create Pivot Table.

Task 41

Merge Employee and Department tables.

Task 42

Concatenate monthly employee files.

9 Final Report

Generate the following report.

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ABC TECHNOLOGIES

EMPLOYEE ANALYTICS REPORT

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Total Employees

Department Wise Employees

Highest Salary

Lowest Salary

Average Salary

Median Salary

Salary Variance

Department Wise Average Salary

Top 5 Performers

Promotion Eligible Employees

Bonus Eligible Employees

Average Experience

Duplicate Records Removed

Missing Values Filled

Files Generated

CSV

JSON

Pickle

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10 Project Folder Structure

Employee_Analytics_Project/

```
data/  
    employees.csv  
    employees.json  
    employees.pkl
```

```
reports/  
    summary.txt  
    analytics.csv
```

```
backup/
```

```
logs/
```

```
employee.py
```

```
analytics.py
```

```
file_handler.py
```

```
utils.py
```

```
main.py
```

```
README.md
```

11 Deliverables

- Clean and modular Python code
- Proper Function Design
- Object-Oriented Programming
- Exception Handling
- NumPy Analytics
- Pandas Data Manipulation
- CSV, JSON and Pickle Storage
- Well-documented README
- Push the complete project to GitHub